ALMA Proposal Preparation Workshop

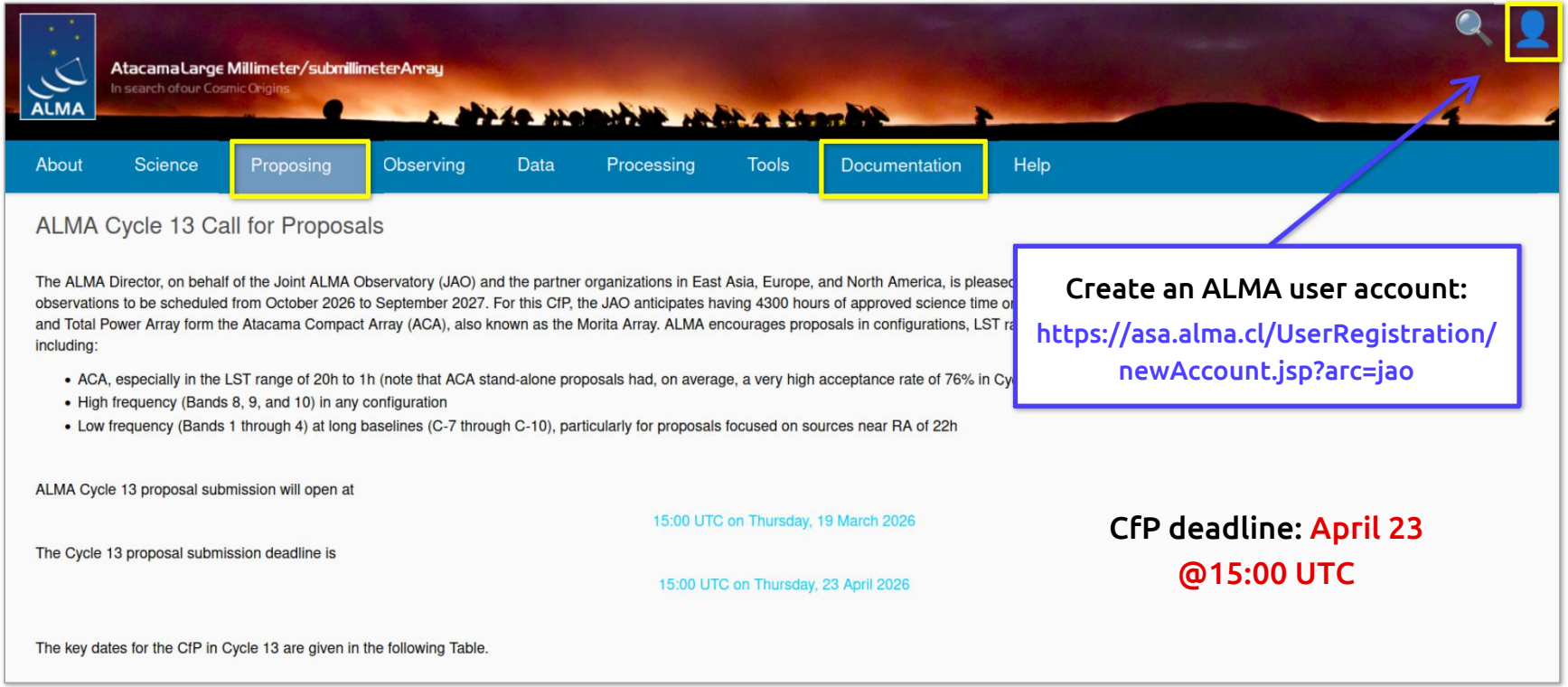
March 20, 2026 @IFA-UV, Valparaíso

2.6. ALMA Observing Tool (OT)

Carla Cornil-Baïotto

carla.cornil@postgrado.uv.cl

Cycle 13 Call for Proposal Portal



Create an ALMA user account:
https://asa.alma.cl/UserRegistration/newAccount.jsp?arc=jao

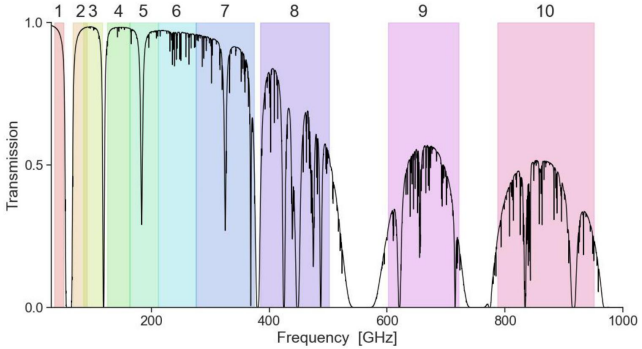
CfP deadline: April 23 @15:00 UTC

Cycle 13 CfP: https://almascience.nrao.edu/proposing/call-for-proposals

[Reminder] ALMA Science & Capabilities

10 receiver bands

(35-950 GHz, $\lambda \approx 0.3 - 9$ mm)

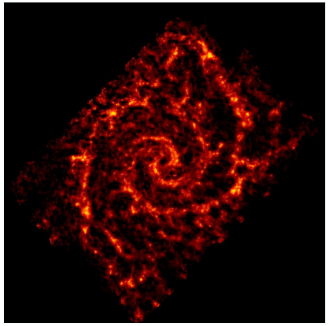


Baselines: 0.16 to 16 km

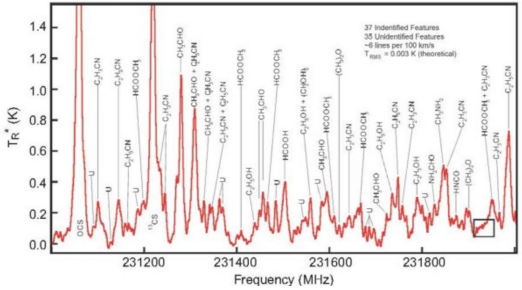
(resolution as fine as 0.005″ at 950 GHz)

Imaging

High-fidelity imaging

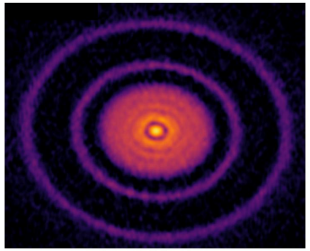


Spectroscopy



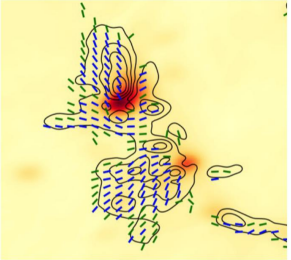
Sensitive spectroscopy to study molecular gas and chemistry

Continuum



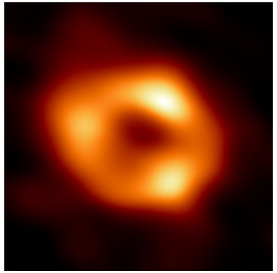
Trace dust and SF

Polarization



Probe magnetic fields

VLBI



EH scale resolution

Slide courtesy of John Carpenter

[Reminder] Cycle 13 Capabilities

Observing mode	12-m Array	7-m Array	TP Array
Spectral line imaging	Bands 1–10	Bands 1 and 3–10	Bands 1 and 3–10
Continuum imaging	Bands 1–10	Bands 1 and 3–10	Bands 1 and 3–10
Spectral scan mode	Bands 1–10	Bands 1 and 3–10	Bands 3–8
New Band 2 (67–116 GHz)	Yes	No	No
Mosaic observations	Bands 1–10	Bands 1 and 3–10	N/A
Polarization	Full polarization (linear + circular), Bands 1–7	Linear only, Bands 1 and 3–7	Not supported
Band-to-band phase calibration (B2B)	Available (Bands 7–10, conditional on calibrator)	Available (Bands 7–10, conditional on calibrator)	Not available

[Reminder] Frequency Sidebands

- ➤ Sky frequencies ~10^2 GHz
  - Too high to be processed directly by the electronics.
  - Sky signal is converted to a lower frequency (IF) using a very stable frequency (LO).

IF ≡ Intermediate Frequency
LO ≡ Local Oscillator

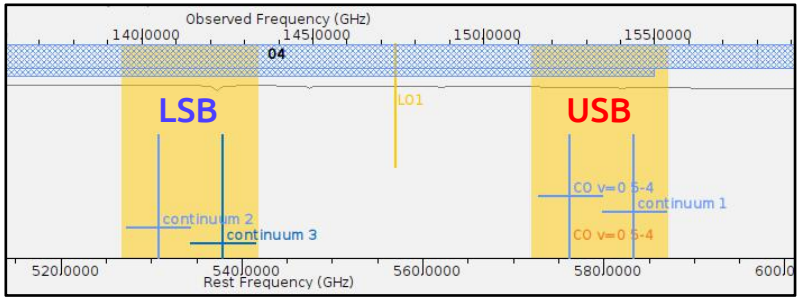
- ➤ Upper & Lower Sidebands
  - → USB = LO + IF
  - → LSB = LO – IF

Heterodyne principle: receiver bands produce two sidebands.

Note: The lower sideband of Band 1 is filtered out, i.e., this is the only receiver with a single sideband.

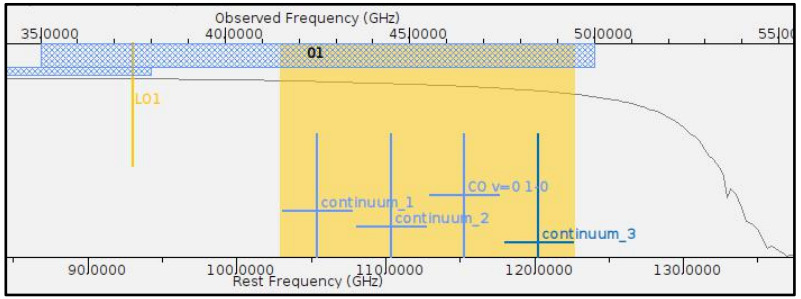
Upper and Lower Sidebands in the ALMA Observing Tool (OT)

Band 4 ➤ (2SB)



Program #2024.1.00015.S, PI Cornil-Baïotto

Band 1 ➤ (SSB)



Program #2025.1.00006.S, PI Cornil-Baïotto

ALMA Observing Tool (OT)

Cosmic Noon Foundry: Peering into the Highly Magnified ISM of the Molten Ring — **2025.1.00006.S**

ABSTRACT

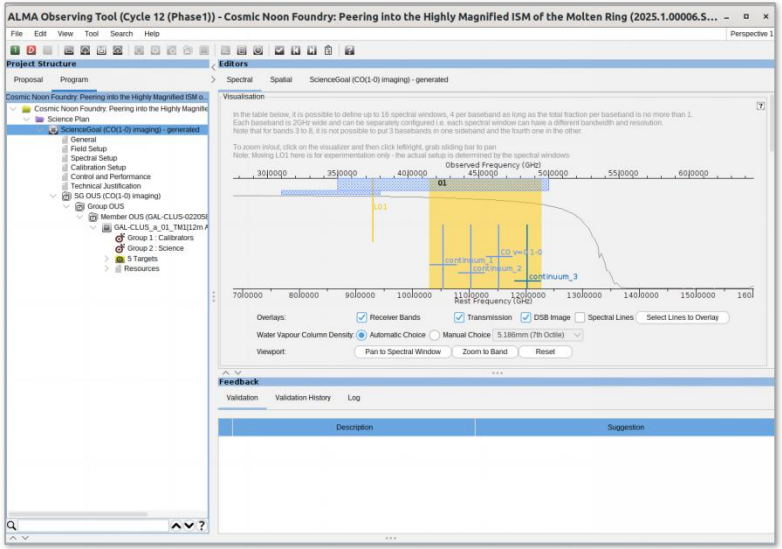
The molecular gas content of galaxies is closely tied to the rates and timescales of star formation. At cosmic noon, galaxies exhibit high molecular gas fractions, resulting in turbulent disks forming dense UV-bright stellar clumps. Resolving the CO and dust emission from their parent giant molecular clouds (GMCs) is essential to understand the physical mechanisms governing star formation. However, sub-kpc observations at these redshifts remain challenging due to the sensitivity and resolution needed. We propose 4.7h Band 1 observations with the ALMA 12m array to image CO(1-0) emission in the Molten Ring galaxy at z~1.5, one of the most extended Einstein rings known. Thanks to strong lensing (20x magnification) and ALMA capabilities, we will probe GMC physical properties on 200-540 pc scales, including their morpho-kinematics, molecular gas masses, depletion times, and star formation efficiencies along with VLT/FORS2 [OII] data. We will test the Kennicutt-Schmidt and Larson's scaling laws at sub-kpc scales and complement similar efforts across cosmic noon, contributing to build a more complete picture of star formation in main-sequence galaxies at a key epoch for galaxy evolution.

SCIENCE CATEGORY:	Galaxies and Galactic Nuclei				
ESTIMATED 12-M TIME:	4.7 h	ESTIMATED 7-M TIME:	0.0 h	ESTIMATED TP TIME:	0.0 h
DUPLICATE OBSERVATION JUSTIFICATION:					

REPRESENTATIVE SCIENCE GOALS (UP TO FIRST 30)

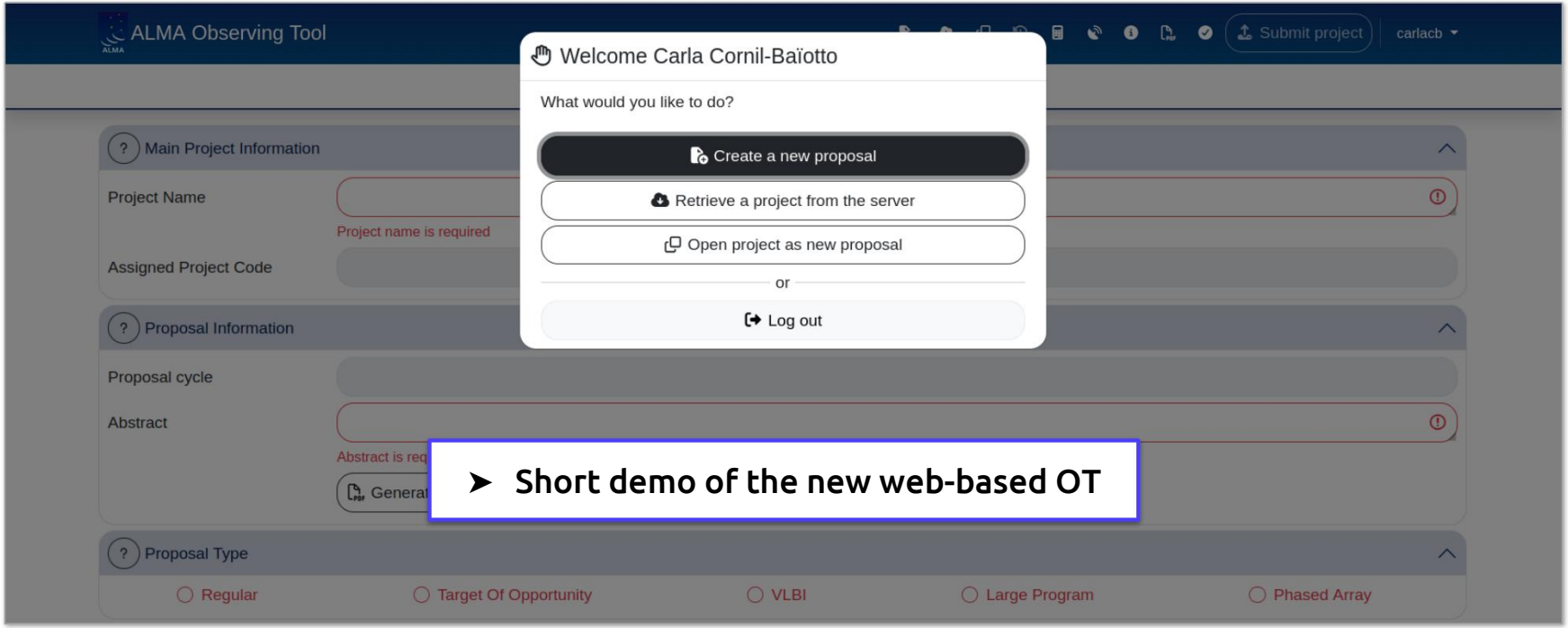
SCIENCE GOAL	CLUSTER	POSITION (ICRS)	BAND	ANG.RES.(")	LAS.(")	12m time (hrs)	7m time (hrs)	TP time (hrs)	Number of sources
CO(1-0) imaging	CO(1-0) imaging	02:20:57, -38:33:11	1	1.600 - 0.600	9.000	4.7	N/A	N/A	1
Total # Science Goals : 1									

SCHEDULING TIME CONSTRAINTS	NONE	TIME ESTIMATES OVERRIDDEN ?	No	JOINT PROPOSAL?	No



➤ **Short demo of the (old) local OT**

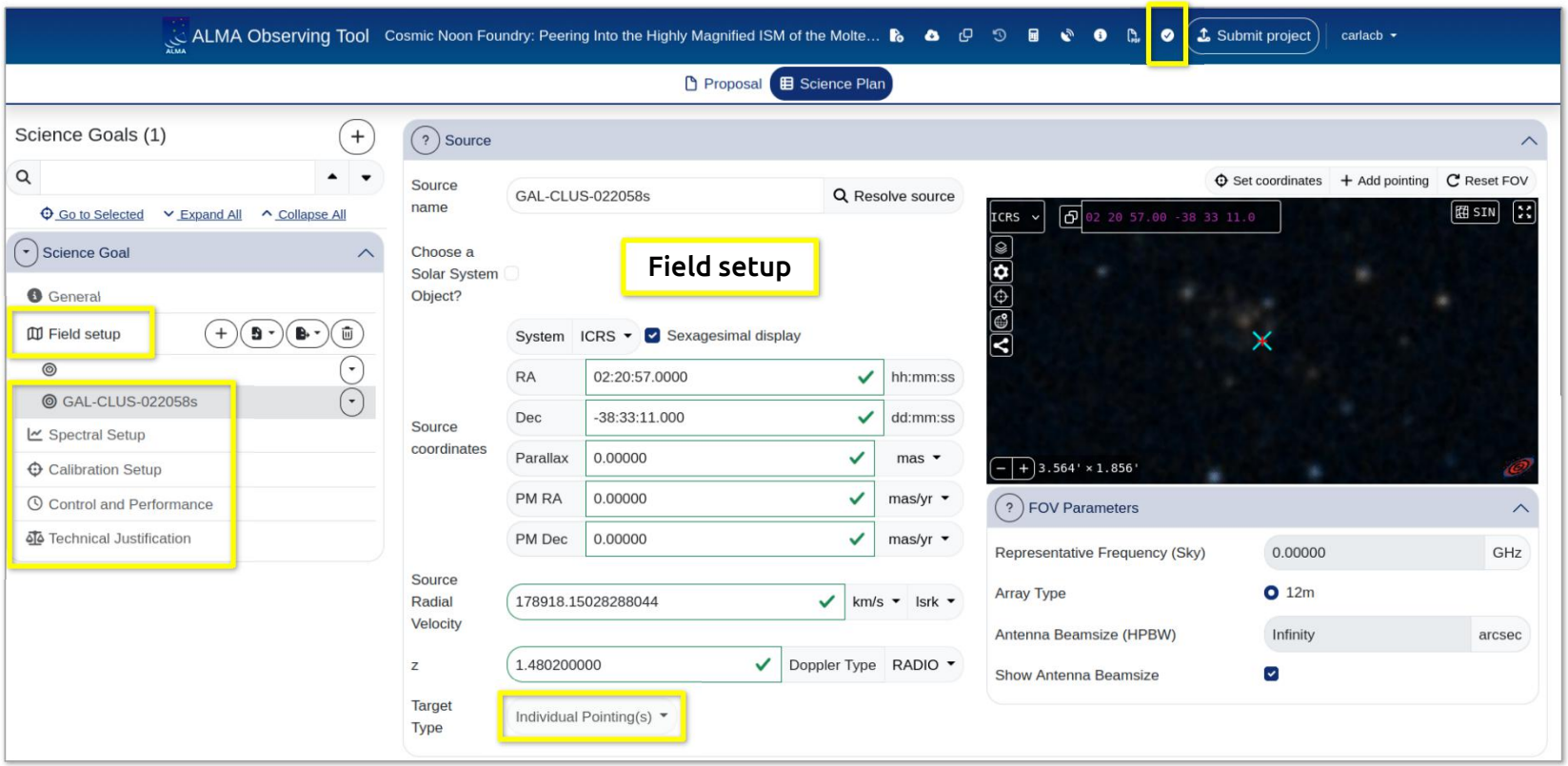
New Web-Based OT



➤ **Short demo of the new web-based OT**

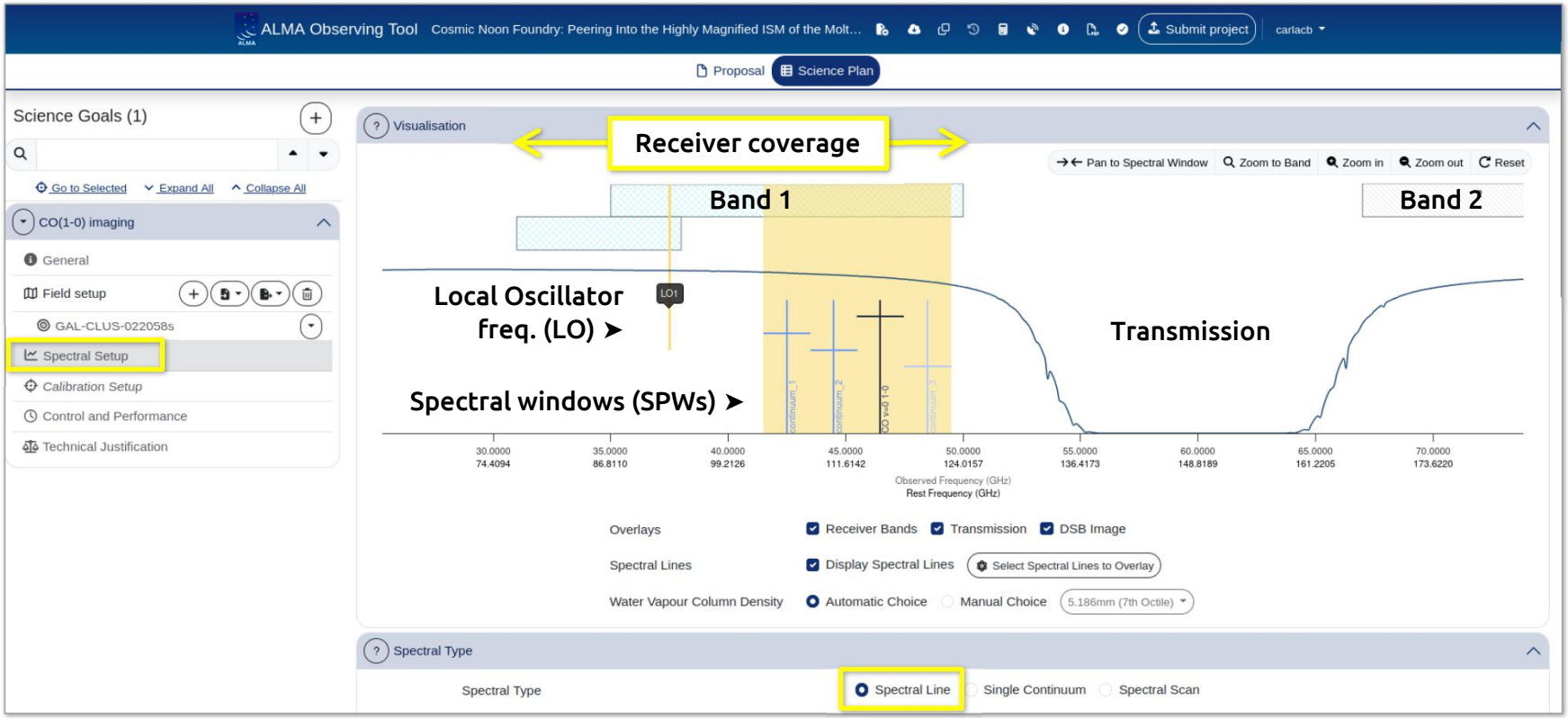
ALMA ngOT: https://cycle-13.sps.alma.cl/ngot/

New Web-Based OT



ALMA ngOT: https://cycle-13.sps.alma.cl/ngot/

New Web-Based OT



ALMA ngOT: https://cycle-13.sps.alma.cl/ngot/

New Web-Based OT

Summary

- **Main differences from classic OT**
  - Web-based only (no local files)
  - Single version (autosave; each save overwrites)
  - Status: Draft / Submitted / Sub. with changes
  - Shared editing (Co-Is/Co-PIs)
- **Known issues/tricks**
  - No proposal templates
  - Switching pointing modes (Regular/Individual) or coords. systems can create many pointings
  - Band 2 continuum setup is limited
  - Loss of mosaic pointings when changing source
  - No “single” angular resolution option

Sources of Support

- **OT video tutorials**

  https://almascience.nrao.edu/proposing/observing-tool/video-tutorials
- **ALMA Helpdesk**

  Questions are usually answered within 48 hours

  Submit a ticket: https://help.almascience.org
- **Face-to-face Visitor Support**

  The NRAO/NAASC offers face-to-face meetings (online or in-person) for proposal preparation support and data reduction (PI data or archival)

  https://science.nrao.edu/facilities/alma/visitors-shortterm

ALMA ngOT: https://cycle-13.sps.alma.cl/ngot/